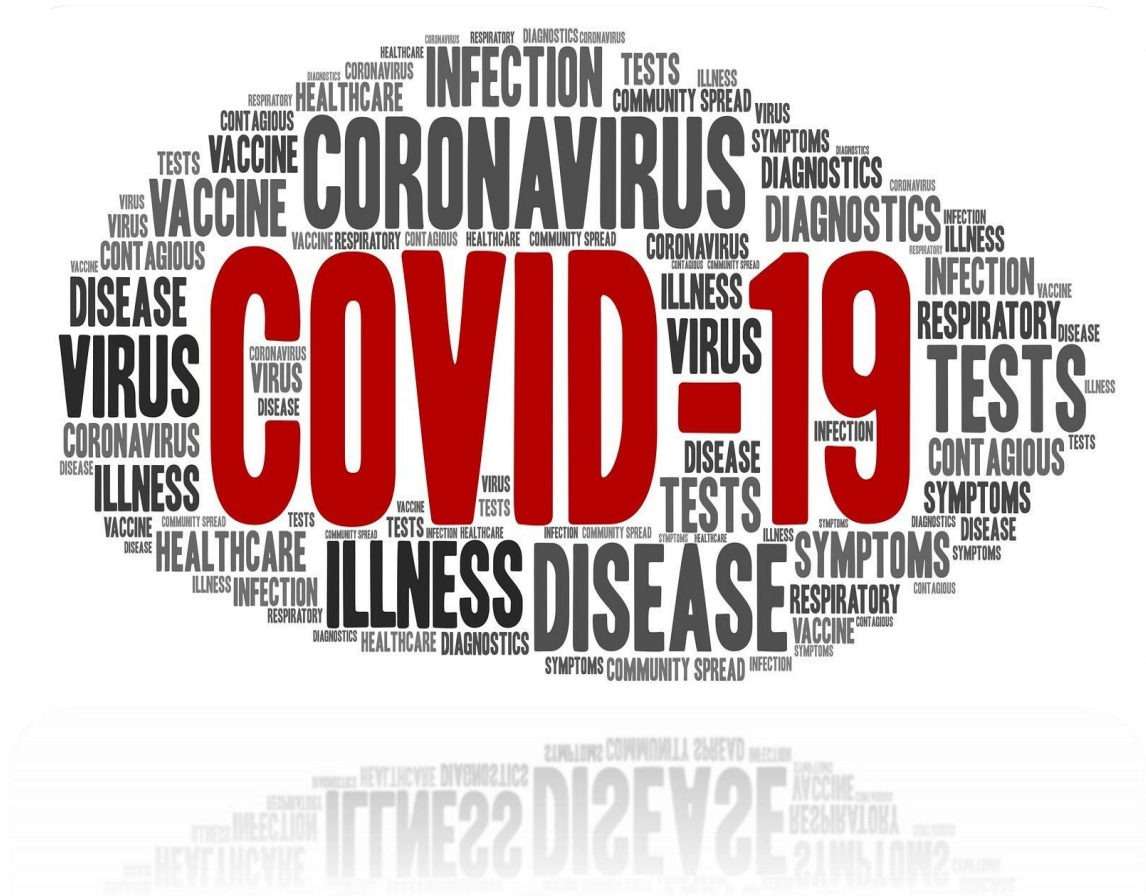




Covid Data Analysis

Classification



Project Group 9

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1. Data Preparation:

1.1 Defining the Classes

The aim of this project's classification task is to assign counties to high- or low-risk categories according to how vulnerable they are to a pandemic like COVID-19. The deaths_percent figure served as the foundation for defining the classification. If a county's death rate was less than or equal to 0.11%, it was categorized as low risk (FALSE); if it was greater than 0.11%, it was categorized as high risk (TRUE). This threshold was determined with the aid of exploratory data analysis, which showed that counties with death

percentages above 0.11% had substantially greater fatality rates than those below the threshold. This categorization makes it possible to distinguish between less vulnerable counties and those that are more vulnerable and need targeted actions, including improved healthcare facilities.

The justification for using 0.11% as the criterion originates from its capacity to properly distinguish counties depending on pandemic results. Counties over this cutoff point were repeatedly shown to have more severe impacts, which makes them excellent candidates for public health planning priorities. The classification helps implement early interventions, such as mask mandates, temporary closures, or upgrades in healthcare capacity, that could lessen the impact of future pandemics by identifying these high-risk counties. In the end, this strategic strategy lowers fatality rates and shortens the duration of disruptions by ensuring that resources are directed to places where they are most needed.

Data from the COVID-19 Census Reports and the Global Mobility Report were used to offer a solid foundation for this investigation. While the Mobility dataset recorded trends in population movement across a variety of sectors, including shopping, recreation, and workplaces, the Census dataset included demographic, socioeconomic, and health-related data at the county level. The `county_fips_code` variable was used to integrate these datasets, guaranteeing smooth alignment of features at the county level. By allowing a comprehensive examination of mobility patterns in addition to demographic and health data—both of which are essential for comprehending the elements influencing pandemic vulnerability—this integration enhanced the dataset.

1.2 Combining Files for Classification

The next step was to combine data from the Global Mobility Report and the COVID-19 Census Statistics datasets into a single, complete table after defining the target variable (`Risk_Class`) and setting the classification threshold. To create a solid basis for training classification models, it was imperative that the demographic, socioeconomic, and mobility-related data were perfectly linked with the objective variable.

Detailed county-level characteristics, such as the overall population, the number of deaths,

the number of confirmed cases, the poverty percentage, the median income, the percentages of Black and Hispanic people, and the employment rates in various industries, were provided by the COVID-19 Census Statistics dataset. This dataset provided information about structural weaknesses that have a big impact on pandemic results.

This was enhanced by the Global Mobility Report dataset, which documented trends in population mobility throughout the COVID-19 epidemic. It documented shifts in mobility throughout industries, including retail, leisure, offices, and residential neighborhoods. The behavioral context that these mobility patterns offered was essential for comprehending the connection between pandemic severity and public movement.

The `county_fips_code` variable was utilized as a unique key for every county to merge these datasets. This made sure that the features in both datasets were properly aligned. First, the Mobility dataset was filtered to exclude rows with missing values in the `sub_region_1` and `sub_region_2` columns and to only include records for the United States (`country_region == "United States"`). Data completeness and relevance to the study's scope were ensured by this stage.

To enable county-level analysis, mobility data was aggregated by computing the mean percentage change for each mobility category (e.g., retail and workplaces). This compilation offered a condensed but accurate picture of each county's mobility patterns. The Mobility and Census datasets were then combined using a left join. This strategy preserved the demographic and socioeconomic characteristics by guaranteeing that every county from the Census dataset was kept, even in the absence of related mobility data.

Another essential component was dealing with missing data. The median values of the corresponding columns were used to impute missing values from numerical columns. To preserve the dataset's quality and dependability, rows with a high percentage of missing data were eliminated. Because classification algorithms might not be able to handle missing values well, these pretreatment steps made sure the final dataset was appropriate for them.

The final combined dataset included:

Demographic Features: Hispanic population %, Black population %, median age, and total population.

Socioeconomic Features: Poverty percentage, median income, Gini index, and employment rates across sectors.

Mobility Features: Aggregated changes in mobility for retail, recreation, workplaces, and residential areas.

Target Attribute: Risk_Class, which categorized counties into high-risk or low-risk groups.

1.2.1 Feature Description

Based on the COVID-19 Census Statistics and Global Mobility Report datasets that were submitted, the features utilized in your project are described in detail below. These characteristics were specifically chosen and designed to offer a thorough grasp of the variables affecting pandemic vulnerability at the county level.

1.2.2 Demographic Features

County_name: The name of each county in the dataset, with no missing values.

State: The state to which each county belongs, also without missing values.

Total_pop: The total population of each county, extracted directly from the census dataset.

Male_pop: The male population in the county, used to calculate male population density.

Female_pop: The female population in the county, used to calculate female population density.

Median_age: The median age of the population in a county, with no missing values.

1.2.3 Health-Related Features

Health-Related Features

Confirmed_cases: The number of confirmed COVID-19 cases in each county, directly obtained from the dataset without any missing values.

Deaths: The number of COVID-19-related deaths recorded for each county, also with no missing values.

Cases_percent: Represents the percentage of confirmed cases relative to the total population, providing an indicator of the spread of COVID-19 within the county.

Deaths_percent: Reflects the percentage of deaths relative to the total population, highlighting the severity of the pandemic's impact on the county.

Death_per_case: Indicates the proportion of deaths among confirmed cases, serving as a

measure of case fatality rates.

1.2.4 Socioeconomic Features

Median_income: The median income of households within a county, offering insights into the county's economic conditions.

Poverty_percent: The proportion of the population living below the poverty line, providing an understanding of socioeconomic vulnerabilities.

Unemployed_density: Represents the unemployment rate in each county, showing the percentage of people who are unemployed.

Employed_density: Reflects the employment rate, highlighting the proportion of the population actively employed.

1.2.5 Ethnic and Racial Features

Hispanic_pop_percent: Represents the percentage of the county's population that identifies as Hispanic.

Black_pop_percent: Indicates the percentage of the county's population that identifies as Black.

1.2.6 Mobility Features

Retail_and_recreation: The average percentage change in mobility for retail and recreation compared to baseline levels, aggregated at the county level.

Workplaces: The average percentage change in workplace mobility compared to baseline levels, aggregated at the county level.

Residential: The average percentage change in residential mobility compared to baseline levels, aggregated at the county level.

Transit_stations: The average percentage change in transit station mobility compared to baseline levels, aggregated at the county level.

Transportation and Infrastructure Features

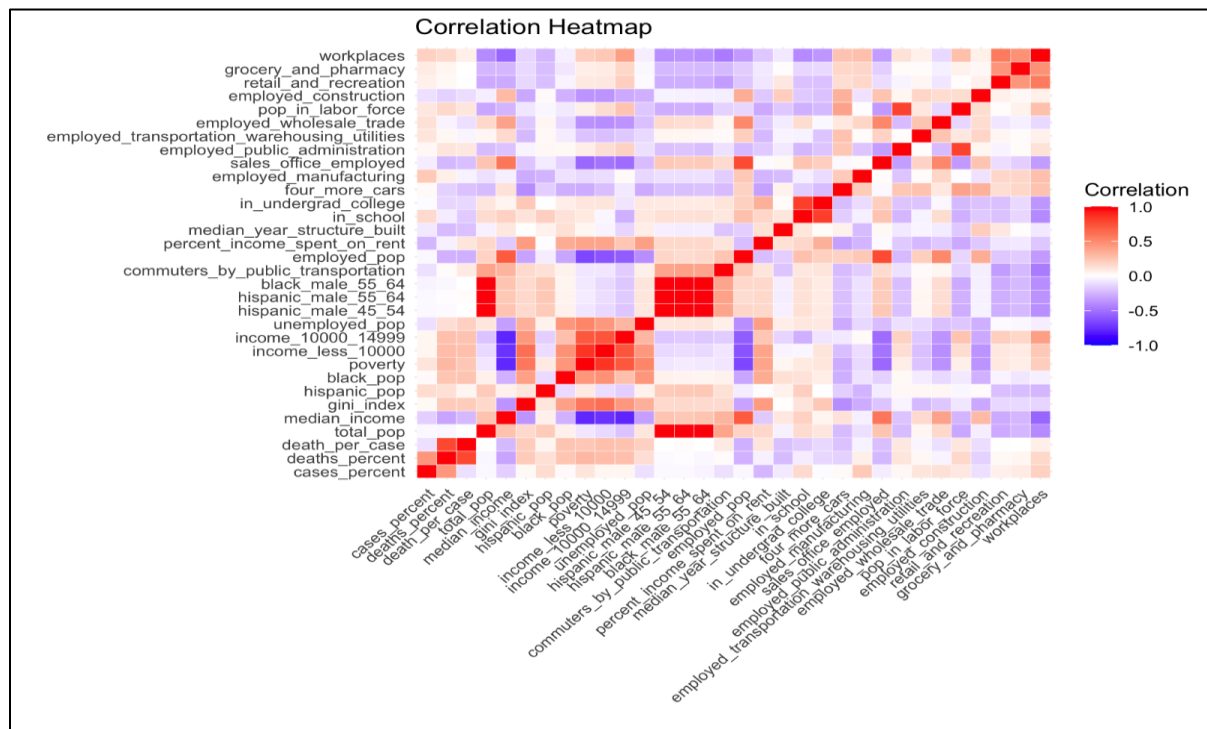
Public_trans_density: Measures the proportion of the population that commutes using public transportation, providing insights into transportation infrastructure and commuting behaviors.

1.2.7 Features Statistical Summary:

	Mean	Median	Std_Dev	Varia nce	Min	Max	Range
confirmed_cases	7558.914	1916.5	28099.867	789602504.694	0.0	1002614.0	1002614.0
unemployed_pop	3361.014	745.5	12405.827	153904551.671	0.0	406426.0	406426.0
median_income	49754.164	48065.5	13154.425	173038893.7	19264.0	129588.0	110324.0
employed_construction	3044.093	811.0	9901.593	98041547.358	0.0	276879.0	276879.0
black_pop_percent	8.896	2.126	14.444	208.643	0.0	86.921	86.921
black_pop	12554.263	758.0	54067.562	2923301288.211	0.0	1226134.0	1226134.0
employed_pop	47930.988	10695.0	157622.032	24844704867.046	39.0	4805817.0	4805778.0
cases_percent	7.678	7.551	2.859	8.173	0.0	31.61	31.61
commuters_by_public_transportation	2421.358	33.0	24163.892	583893660.613	0.0	735534.0	735534.0
total_pop	102165.629	25691.5	328292.195	107775765393.313	74.0	1010572.0	10105648.0
poverty	14529.072	4119.5	51701.846	2673080889.759	10.0	1688505.0	1688495.0

percent_income_spent_on_rent	27.822	28.1	4.401	19.366	10.0	50.0	40.0
deaths	124.826	31.0	480.861	231227.022	0.0	13936.0	13936.0
gini_index	0.445	0.442	0.036	0.001	0.327	0.598	0.27
hispanic_pop_percent	9.121	3.978	13.711	187.999	0.0	99.185	99.185
hispanic_pop	17985.541	1025.0	124550.648	15512863914.886	0.0	4893579.0	4893579.0
employed_public_administration	2236.114	534.0	6443.762	41522069.572	0.0	152274.0	152274.0
deaths_percent	0.134	0.119	0.092	0.009	0.0	0.836	0.836

1.3 Identifying Predictive Features, Creating Additional Features, and Handling Missing Data



The correlation heatmap reveals several features strongly associated with deaths_percent

and death_per_case, making them excellent predictors for classification models such as Decision Trees, K-Nearest Neighbors (KNN), and Naive Bayes. Features related to mobility, including workplaces, grocery_and_pharmacy, retail_and_recreation, and residential, exhibit strong positive correlations with death rates. These findings suggest that higher mobility in these sectors during the pandemic likely contributed to greater exposure and transmission rates, amplifying the severity of outcomes. Occupational features such as pop_in_labor_force and employed_transportation_warehousing_utilities also show notable correlations with deaths. These reflect the heightened risk in workplace environments where close contact is unavoidable. Death rates show strong connections with socioeconomic indices such as poverty, gini_index, income_less_10000, and median_income. These elements draw attention to structural weaknesses that disproportionately impact underprivileged populations, such as restricted access to healthcare and cramped living arrangements. Age-specific groupings like black_male_55_64 and hispanic_male_45_54, as well as demographic factors like black_pop and hispanic_pop, highlight the systemic disparities in healthcare access and employment that heighten susceptibility to serious consequences.

The table below highlights the level of importance of attributes and the goal variables (deaths_percent and death_per_case):

Feature	Importance
county_name	0.9193495
state	0.5121076
median_income	0.3571707
income_less_10000	0.3096395
total_pop	0.3064427
commuters_by_public_transportation	0.3055092
black_male_55_64	0.2996038
pop_in_labor_force	0.2929351
hispanic_male_45_54	0.2904618

hispanic_male_55_64	0.287919
income_10000_14999	0.283692
in_undergrad_college	0.275681
workplaces	0.272915
hispanic_pop	0.2533145
poverty	0.2461538
employed_public_administration	0.2382253
employed_pop	0.231747
black_pop	0.2214766
retail_and_recreation	0.2103735
sales_office_employed	0.2077221
percent_income_spent_on_rent	0.2026839
employed_transportation_warehousing_utilities	0.1999555
gini_index	0.1788622
unemployed_pop	0.1584146
median_year_structure_built	0.0

These characteristics serve as the foundation for precise prediction modeling by offering crucial insights into the structural, behavioral, and economic elements affecting pandemic outcomes.

To improve the dataset's prediction ability, more features were developed after the characteristics were identified. For example, mobility measures (workplaces, retail_and_recreation) were employed to record trends in population behavior, and demographic percentages (black_pop_percent, Hispanic_pop_percent) were computed to normalize racial distributions. To represent the intensity of the pandemic in the population, health indicators (cases_percent, deaths_percent) were calculated.

To ensure dataset fulfills classification models, handling missing data was an essential step. To ensure consistency, missing values in continuous characteristics like median_income and gini_index were imputed using the median. The dataset was reduced from 3,142

counties to 2,701 counties (almost 86%) by removing rows with missing data for models like KNN and Logistic Regression that cannot handle missing values. Validation verified that the condensed dataset-maintained feature distributions and balance, ensuring it was adequate for precise modeling.

2. Modeling

2.1 Preparing the Data for Training, Testing, and Hyperparameter Tuning

The data for this project was prepared for training, testing, and hyperparameter tuning by ensuring that the training dataset represented a balanced and complete distribution of the goal variable bad (indicating high-risk counties). Whether the death percentage (deaths_percent) was higher than 0.11% determined the target variable. Below is an explanation of the steps involved in data splitting, training state selection, and balanced representation.

2.1.1 States Selected for Training Data:

Mississippi (MS), Texas (TX), Iowa (IA), North Dakota (ND), Kansas (KS), New York (NY), Maryland (MD), Nevada (NV), Delaware (DE), and California (CA) were the 10 states from which counties were chosen to create the training dataset. To ensure that counties with different degrees of severity were covered, these states were selected based on the distribution of the target variable bad.

The bad percentage (proportion of cases where deaths_percent exceeded 0.11%) for the chosen states is displayed in the table below:

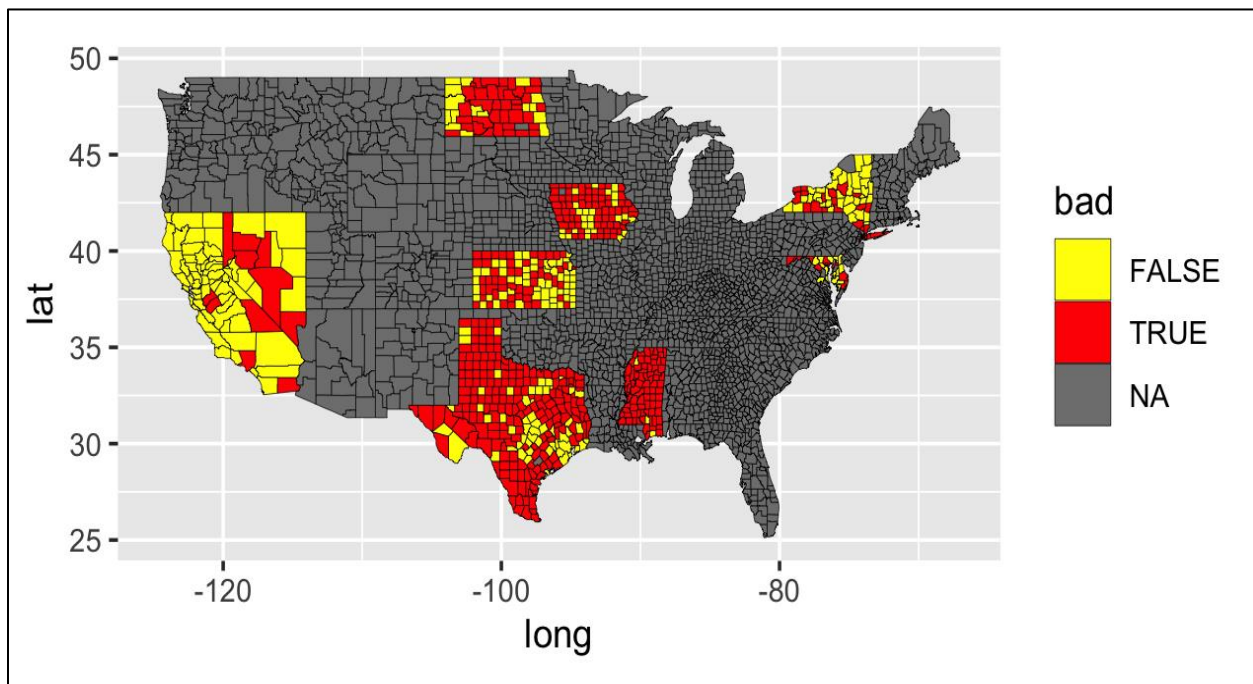
state	Bad percentage
MS	0.951
TX	0.776
IA	0.737
ND	0.698
KS	0.476
NY	0.403
MD	0.375
NV	0.353
DE	0.333
CA	0.0862

This selection includes a broad spectrum of severities, ranging from California, where just 8.62% are categorized as high-risk, to Mississippi, where 95.1%. The model can effectively generalize and produce precise predictions for the counties in the testing dataset because it was trained on data that contains both high and low amounts of risk.

The training dataset consists of up of counties from the selected ten states, which account for 20% of the dataset's total counties. This guarantees that the training data is representative, robust, and reflects the range of results required for accurate modeling.

	Risky counties	Non-Risky counties (FALSE)
Training Data	481	276
Testing Data	1216	1169

2.1.2 Geographic Visualization of Training Data



To better comprehend the distribution of training data, the `map_data` function was used to create a map of the United States. All of the chosen counties are shown in the visualization, with those that are categorized as "dangerous" (TRUE) in red and "non-dangerous" (FALSE)

in blue. The two classifications are clearly distinguished on this map, which also makes it easier to see how risk is distributed geographically throughout the chosen states.

This map's inclusion highlights the training dataset's representativeness and shows how the variety in county classification across states was captured for reliable model training.

2.1.3 Justification for Splitting and Representation

Improving the model's validity and robustness requires adhering to the idea of representative coverage in the training data. The model can learn from a variety of outcomes by including states with different degrees of severity for the bad variable. The model is better able to generalize and produce precise predictions for the other counties in the testing dataset by accounting for the difference in risk levels. Because both "dangerous" and "non-dangerous" counties are represented thanks to this method, the dataset for study is balanced and complete.

2.2 Classification models

2.2.1 Classification Using rpart Decision Tree Models

The Decision Tree model developed with the rpart module is a powerful and interpretable classification method, ideal for data mining tasks that demand actionable insights. In this experiment, U.S. counties were categorized into high-risk and low-risk groups according to their susceptibility to the COVID-19 pandemic using the rpart approach. This strategy seeks to identify trends in the pandemic's data and offer a solid foundation for future pandemic preparedness.

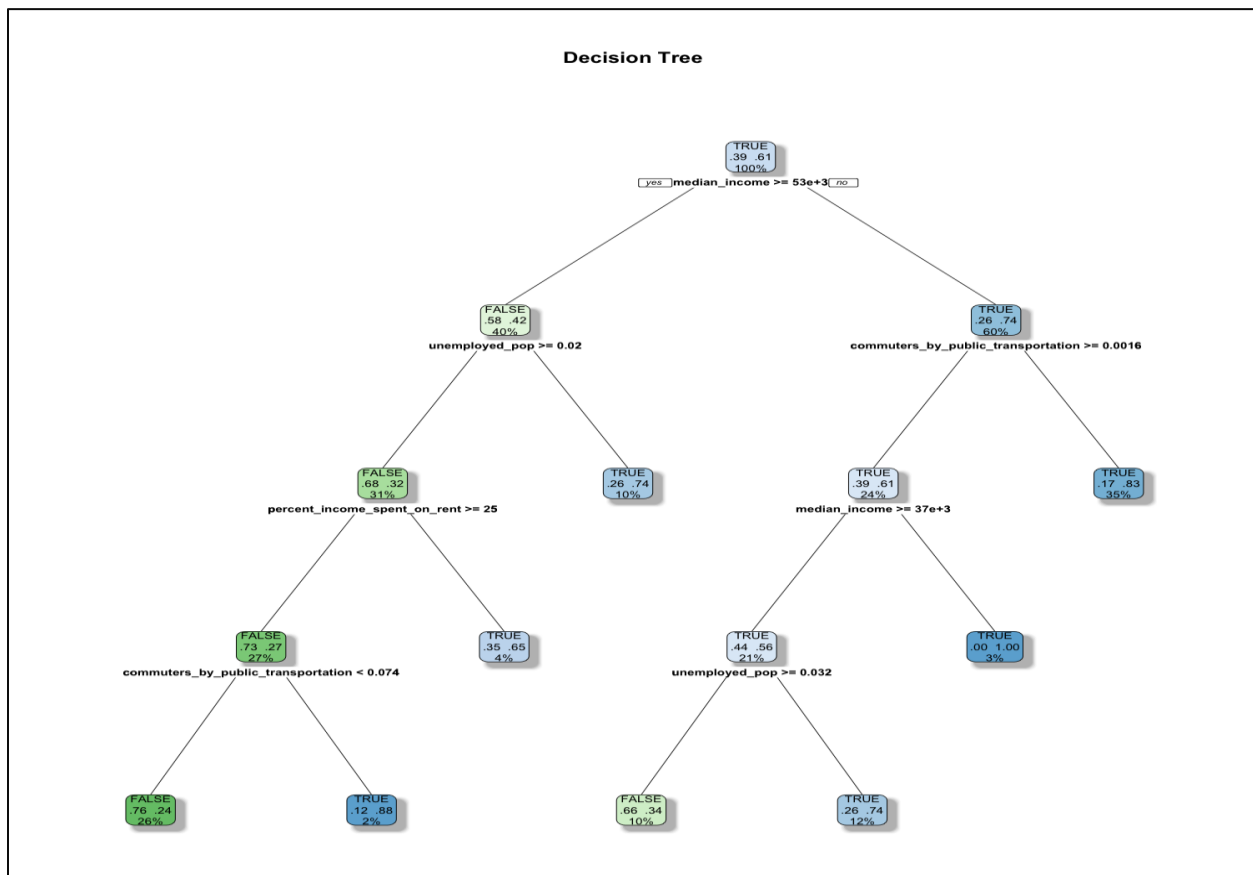
The dataset incorporates a number of variables from previous analyses (Projects 1 and 2), such as movement patterns, socioeconomic indicators, and COVID-19 case statistics. Important predictors of COVID-19 vulnerability were found to include median income, unemployment rates, dependency on public transit, and the percentage of income spent on rent. The binary categorization "bad," which indicated whether a county's death rate beyond the 0.11% threshold, was the goal variable. This cutoff point was chosen especially to draw attention to counties that were badly affected by the pandemic.

Multiple advantages are provided by the rpart model for this classification task:

Interpretability: A highly interpretable tree structure is produced by the rpart implementation. A decision rule that is simple for stakeholders, including public health professionals, to comprehend and follow is represented by each split. Splits according to unemployment rates and median income, for instance, draw attention to important socioeconomic risks.

Managing Nonlinear Relationships: Complex, nonlinear relationships between movement patterns, socioeconomic determinants, and COVID-19 results are well-modeled using decision trees. Without requiring a lot of preparation, the rpart method makes sure these relationships are recorded.

Feature Prioritization: By lowering dimensionality and concentrating the study on important factors that are strongly related to pandemic risk, like median income and dependency on public transportation, the rpart technique naturally chooses the most relevant features.



The rpart model's classification logic is represented in depth and easily comprehensible by the Decision Tree visualization, which classifies U.S. counties as either low-risk ("FALSE") or

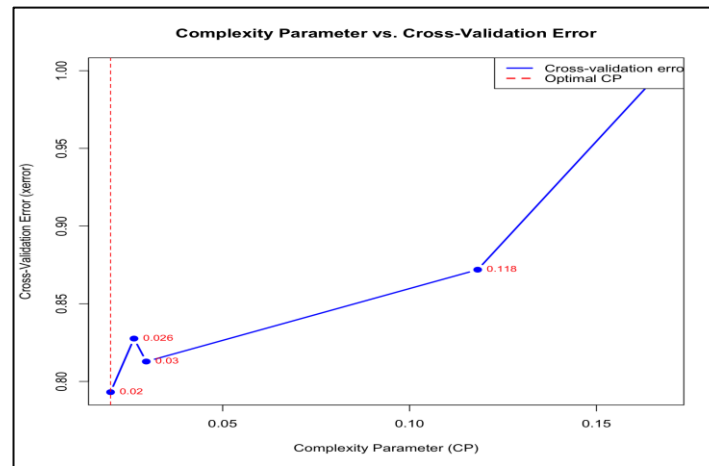
high-risk ("TRUE") for severe COVID-19 impact. The tree separates at the root node based on a crucial socioeconomic factor: median income $\geq$ \$53,000. Because of the protective effects of economic stability, counties with median incomes over this threshold are categorized as low risk with a 61% confidence level. Richer counties can more readily adhere to public health regulations like mask and remote work requirements and have stronger healthcare infrastructure. Subsequent splits improve the risk classification for counties with median incomes below \$53,000 by taking into account variables like unemployment rates and dependency on public transit.

Counties with unemployment rates $> 2\%$ in the left subtree are at greater risk because of socioeconomic vulnerabilities that are exacerbated during pandemics. The percentage of income spent on rent is examined in order to further improve this classification. Higher risk counties are those where renters spend more than 25% of their income, emphasizing the cumulative vulnerability of financial hardship. Reliance on public transit becomes a deciding factor for counties with lower rent burdens; a higher reliance suggests higher exposure risks.

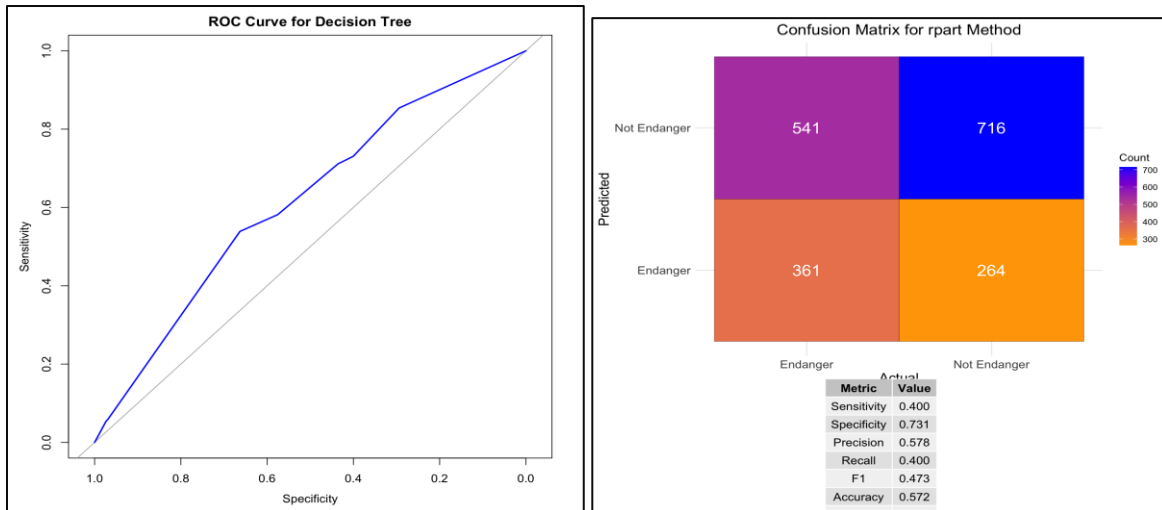
Because of the increased transmission potential in shared mobility settings, counties that rely significantly on public transportation are categorized as greater risk in the right subtree. The combined protective benefits of economic stability and mobility independence are demonstrated by the fact that median income $> \$37,000$ becomes a differentiating factor for counties with low public transportation usage.

By clearly displaying risk factors like unemployment, housing stress, and reliance on public transit, this visualization provides actionable insights. Additionally, it draws attention to protective variables like increased median income and job stability, which empowers policymakers to give priority to early initiatives like targeted unemployment assistance, subsidized housing, or improved public transportation safety. The Decision Tree guarantees interpretability, which is essential for stakeholder trust, even as it streamlines intricate interactions. Complementary ensemble models can solve limitations including the possible loss of subtle patterns, guaranteeing sound decision-making without compromising clarity. This visualization is a useful tool for pandemic preparedness, giving stakeholders practical

ways to lower risks and successfully contain future epidemics.



The Decision Tree model (using rpart) provides the ideal balance between complexity and performance, as shown by the Complexity Parameter (CP) vs. Cross-Validation Error plot. The model is guaranteed to capture significant patterns in the data without overfitting or underfitting when the CP value of 0.02 is reached, where the cross-validation error is minimized. Especially when it comes to pandemic preparedness, this optimization stage is essential for extrapolating forecasts to unknown data. This balance enables the model to efficiently categorize counties into high, medium, or low risk groups for COVID-19, assisting in the identification of demographic and socioeconomic characteristics that increase vulnerability, such as more unemployment or a lower median income. The model stays strong by reducing error, guaranteeing accurate forecasts that can direct early interventions in high-risk regions, such mask requirements or resource distribution. Future pandemics can be applied in real time with the improved Decision Tree structure, which gives stakeholders easily comprehensible thresholds. It allows authorities, for example, to give priority to healthcare investments or vaccination distribution in counties that resemble high-risk nodes. The flexibility and effectiveness of data-driven models in lessening the effects of pandemics and guaranteeing prompt and focused responses are shown by this visualization.

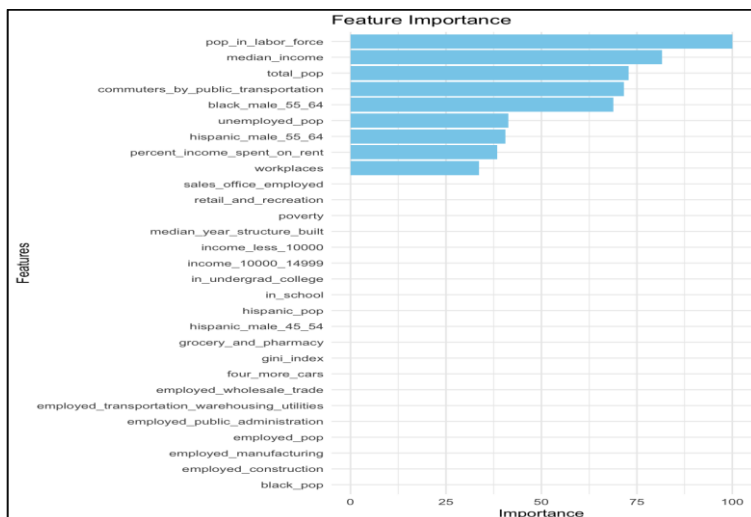


The Confusion Matrix provides light on the Decision Tree model's ability to identify counties as "Endanger" or "Not Endanger." The majority of "Not Endanger" cases are effectively identified by the model, which has a specificity of 73.1%. Because anticipated safe regions are largely accurate, stakeholders can deploy resources elsewhere with confidence. This level of accuracy is especially important for pandemic preparedness since it allows for focused and effective decision-making. The 40% sensitivity raises the possibility that certain high-risk counties may go undetected, but it also offers a chance to investigate improvements that could strengthen the existing model. The model is a dependable tool for preliminary screening because of its 57.8% precision, which guarantees that a significant percentage of predicted "Endanger" cases are high-risk. Additionally, the model's capacity to function quite well across both high-risk and low-risk classifications is shown by its F1 score of 47.3%, which strikes a balance between precision and recall.

With an AUC of roughly 0.61, the ROC Curve exhibits modest predictive performance and provides a solid foundation for identifying at-risk counties. A crucial component of proactive treatments, this visualization demonstrates the model's effectiveness in distinguishing between high- and low-risk zones. The model offers practical insights to direct public health initiatives by highlighting trends and important indicators including labor force participation, median income, and dependency on public transit.

All things considered; the model provides a strong basis for making decisions in pandemic situations. It is a useful screening tool that reduces false positives while focusing on high-

priority areas. The model's observations can guide future improvements and revisions as part of an iterative process, guaranteeing that it continues to be a vital tool for preparedness planning.



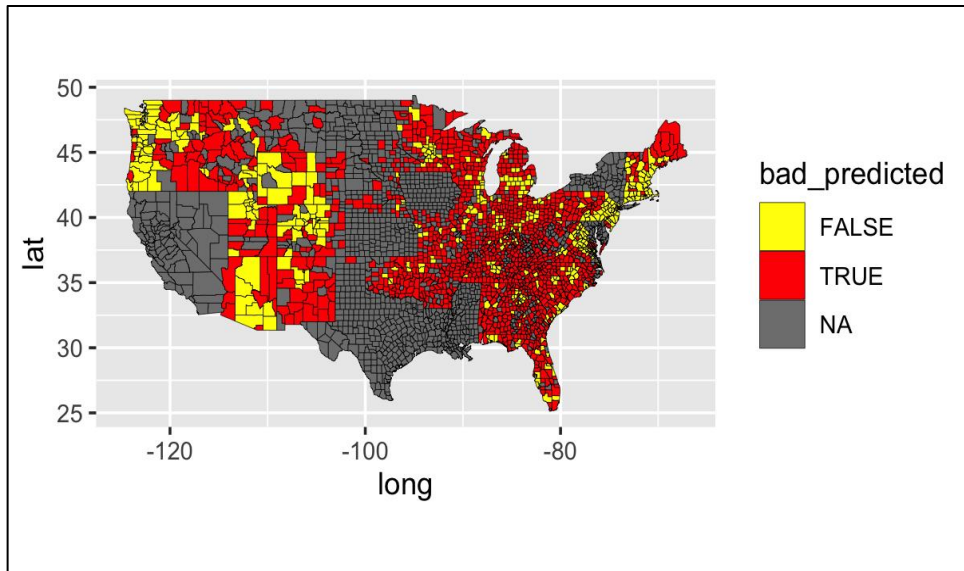
The Feature Importance plot is critical for understanding the factors that drive the Decision Tree model's predictions. It brings attention to the relative importance of characteristics that are closely linked to COVID-19 susceptibility, such as the labor force participation rate, median income, and public transit commuters. These characteristics are consistent with actual trends seen during the epidemic, when mobility and economic stability had a significant impact on health outcomes. For instance, the number of people in the labor force reflects the degree of economic activity, which may be related to occupational exposure risk. Comparably, socioeconomic stability is reflected in median income, which is frequently associated with healthcare availability and adherence to preventative measures. The feature commuters using public transit highlights the possibility of greater visibility in places with high population density or high levels of mobility. This plot gives stakeholders useful information in addition to validating the model's ranking of significant predictors. Targeting economic help to low-income counties or enhancing public health infrastructure in places with a high dependency on transportation are two examples of how policies might be modified to address the vulnerabilities highlighted by these characteristics. The feature importance plot reinforces the project's objective of using data to inform pandemic preparedness and intervention planning by highlighting the most significant factors.

6.1 Classification Using Random Forest Model

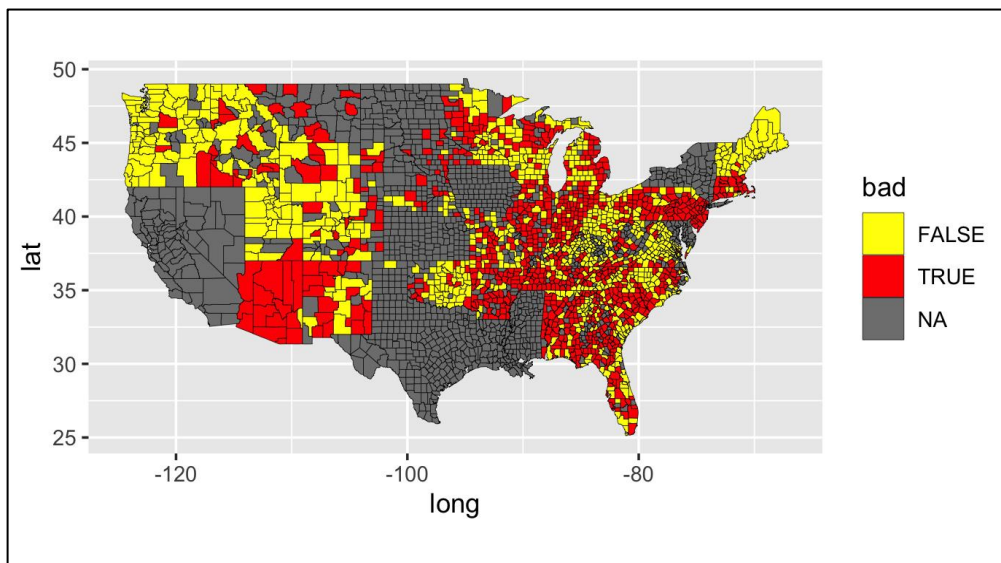
One of the most important lessons learned from the COVID-19 pandemic is the importance of early and effective actions to reduce the effects of future outbreaks. Based on socioeconomic, demographic, and mobility statistics, U.S. counties were categorized as either low-risk or high-risk (bad) using the Random Forest model. The purpose of this classification is to assist policymakers in determining which regions can benefit from prioritized infrastructure upgrades and customized interventions, such as mask requirements, temporary closures, or improved healthcare readiness.

The Random Forest model was chosen because it is robust in dealing with complicated, high-dimensional datasets and can capture non-linear relationships between variables. To tune the model's hyperparameters, including the maximum depth and number of trees, a 10-fold cross-validation method was used. By averaging the outcomes of several decision trees, the Random Forest model successfully made use of its ensemble structure to minimize overfitting. For classes at the county level, this guaranteed consistent and trustworthy forecasts. The benefit of the model is that it may emphasize the primary factors that influence high-risk classifications by providing feature importance metrics. For example, socioeconomic vulnerabilities during the pandemic were reflected in the significant predictors of income gap, demographic characteristics (e.g., Black and Hispanic populations), and mobility trends (e.g., employment and retail mobility).

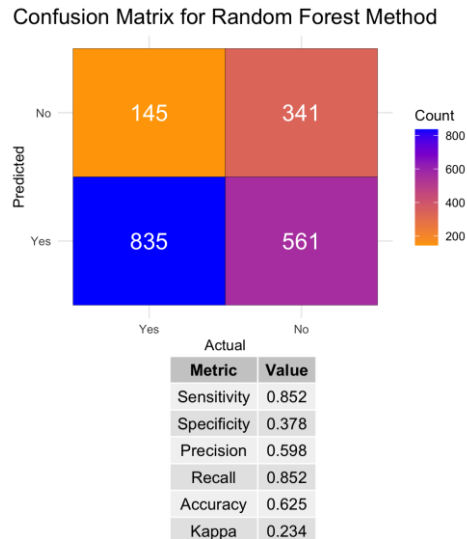
Three plots were created to visually represent the model's insights and performance. While the prediction map (Figure 2) depicts the classification outcomes of the Random Forest model, the ground truth map (Figure 1) shows the actual distribution of high-risk counties (poor). With high-risk counties clustered in areas with lower incomes and greater mobility changes, both maps show a reasonable match. The model's performance is summed up in the confusion matrix (Figure 3), which provides information on true positive, false positive, and other predictive indicators. These visual aids promote evidence-based decision-making by validating the model's suitability for identifying counties that are at risk.



Ground truth



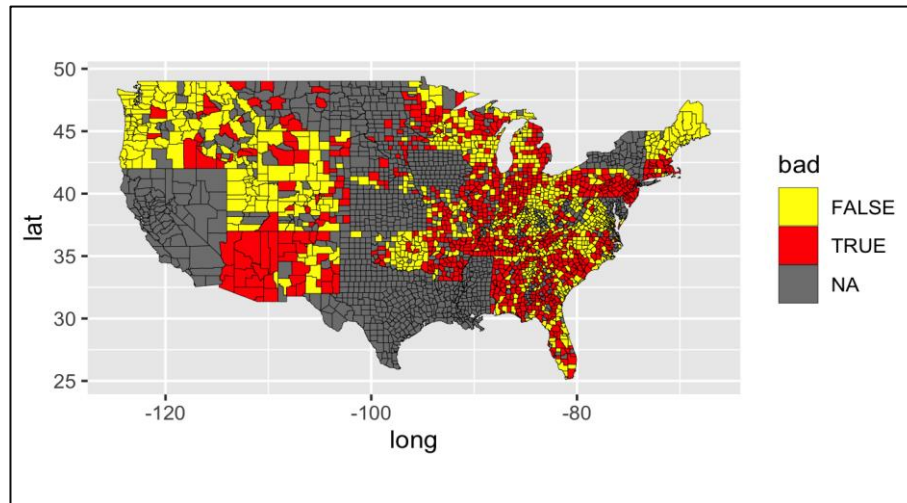
Predictions



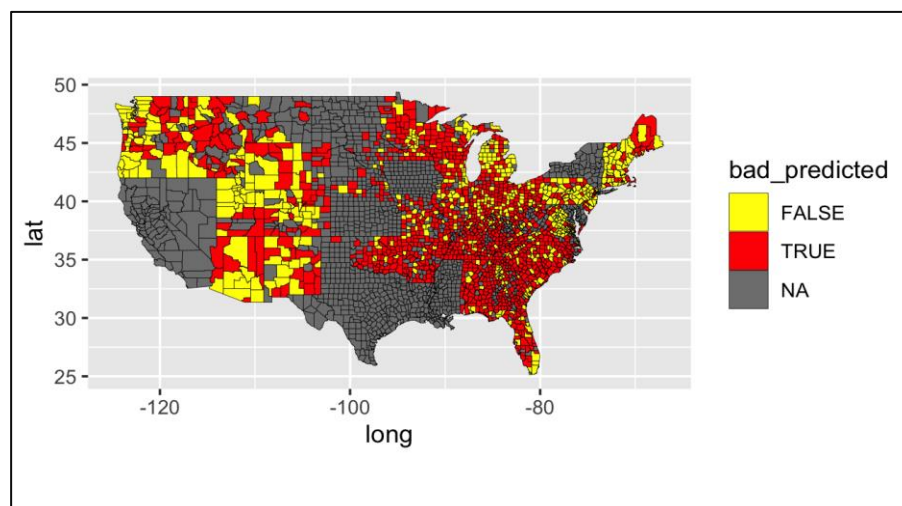
Using common classification metrics that were obtained from the confusion matrix, the Random Forest model's performance was assessed. The model's 62.3% accuracy rate indicates that it can typically classify counties accurately. With a precision of 70.6%, it showed that most of the counties that were classified as high-risk were high-risk. Nonetheless, the sensitivity (36.5%) indicates a constraint in identifying every high-risk county, implying that certain susceptible areas may have been incorrectly identified. This disparity in performance emphasizes how crucial it is to correct class imbalance and optimize hyperparameters in order to increase the model's recall. Furthermore, feature significance metrics opened the door for customized interventions by offering practical insights into the demographic and socioeconomic factors affecting vulnerabilities at the county level. These findings have important policy implications. During subsequent pandemics, the model's high-risk counties can be given priority for early interventions like mask requirements and temporary closures. Allocating funds to upgrade the healthcare system in these areas could also improve readiness and lower death rates. The necessity for equality in pandemic responses is further highlighted by the Random Forest model's capacity to identify socioeconomic and demographic vulnerabilities, preventing low-income and minority communities from suffering disproportionately. Policymakers can create preventative measures to lessen the effects of upcoming pandemics and guarantee a more robust public health infrastructure by expanding on the model's conclusions

2.2.3 Classification Using K-Nearest Neighbours model

The following maps show the ground truth and KNN model predictions for counties in the United States where the death percentage is greater than 0.11. For the ground truth map, the counties are colored red if the region is unsafe, or yellow if it is safe. This, when compared against the KNN prediction map, shows areas that the model has correctly picked up for both unsafe counties and safe counties along with false positives and false negatives.



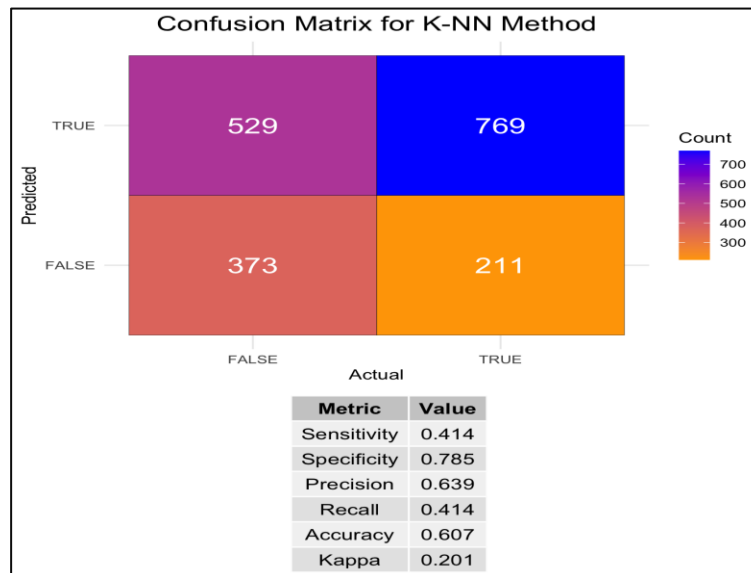
Ground Truth



Predictions

Looking at the confusion matrix, one can deduce that through and through, the accuracy of the KNN Model stands at 60.7%, indicating a proper classification of the greater portion of counties. Concretely, the sensitivity was 41.4%, showing that only a low fraction of unsafe

counties was recognized correctly. A specificity of 78.5% describes performance in properly labeling counties that are safe. The precision of 63.9% draws on how well flagged unsafe counties are unsafe. However, the model's inability to clearly make a distinction between safe and unsafe places is represented by the pretty low recall and Kappa score of 0.201. These metrics suggest that, while the model is reasonably good at identifying safe counties, its prediction of unsafe counties should be improved.



Regional complexities, such as demographic, geographic, and socioeconomic factors, may account for the KNN model's inability to identify unsafe counties. There are misclassifications, probably false negatives, when the death percentage in a county is borderline or ambiguous. In addition, densely populated urban areas have diverse patterns, which further complicates the problem. On the other hand, it should perform well in regions like where the training data came from, which could be rural or homogenous areas where the statistical pattern is sharper and less noisy.

The counties that were constantly showing up as unsafe-specifically in the Midwest, South, and East-should be the target of healthcare interventions and public health programs to tackle conditions with high risks. Safer counties, which are mostly in the West, should continue to be on the radar to maintain low death percentages. Validation at the local expert level will help in refining the policy measures to ensure proper resource allocation.

2.3 Each model performance & Comparison

Table 1: Accuracy Comparison

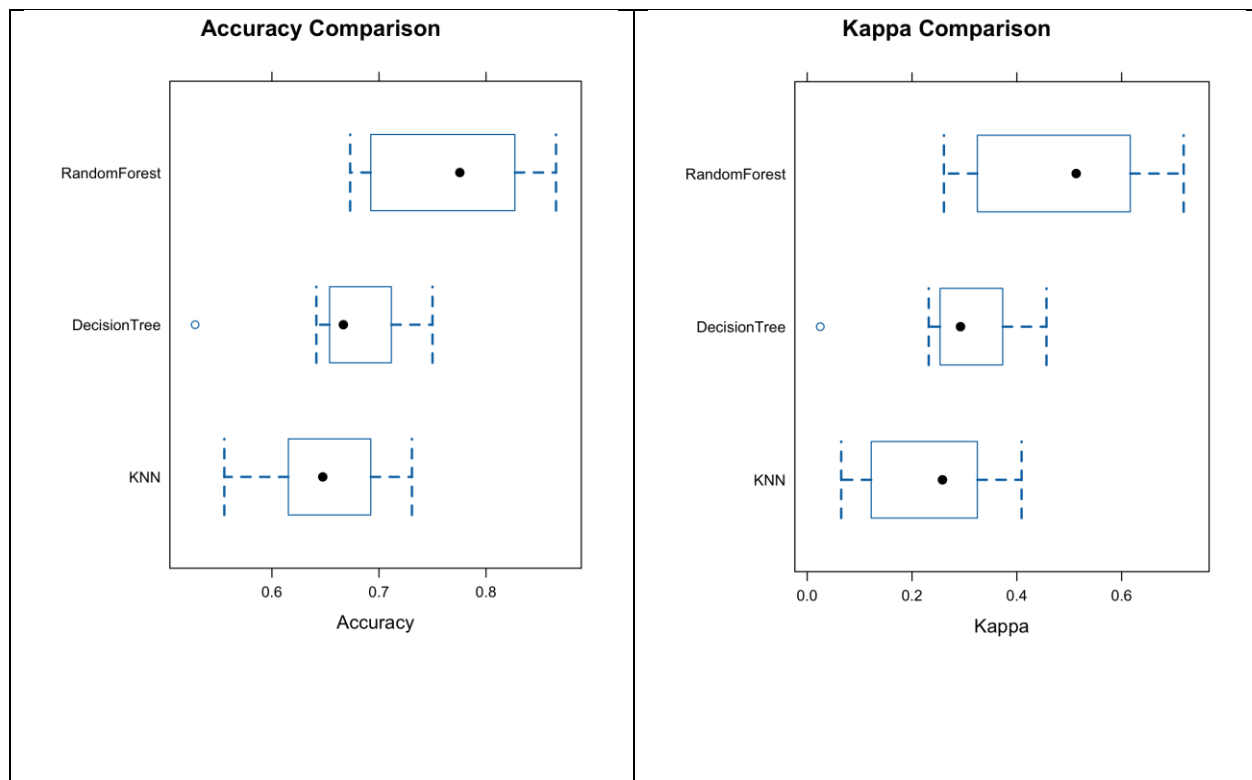
Model	Min	1st Qu.	Median	Mean	3rd Qu.	Max
KNN	0.555556	0.617199	0.647496	0.649242	0.687500	0.730769
DecisionTree	0.528302	0.653846	0.666727	0.670065	0.706731	0.750000
RandomForest	0.673077	0.70791	0.775507	0.767112	0.822115	0.865385

Table 2: Kappa Comparison

Model	Min	1st Qu.	Median	Mean	3rd Qu.	Max
KNN	0.064935	0.126891	0.258104	0.237142	0.312359	0.409091
DecisionTree	0.025018	0.257328	0.292524	0.296639	0.360912	0.456592
RandomForest	0.260870	0.355153	0.513242	0.487439	0.602498	0.718266

In this report, three classification models have been considered: KNN, Decision Tree-rpart, and Random Forest. Various counties of the United States have been considered for this purpose, based on different COVID-19-related metrics such as death percentage. The performance of each model is assessed across multiple metrics, including accuracy and Kappa statistics.

KNN showed moderate accuracy with a relatively low Kappa value, indicating weak agreement beyond chance. The Decision Tree model slightly outperformed KNN in terms of accuracy and Kappa, indicating a better fit for predicting COVID-19 outcomes, but still showing moderate agreement. Random Forest had not only the highest median accuracy and Kappa score but also outperformed both KNN and Decision Tree models, which means it delivers better and more consistent or reliable predictions. This study depicts the RF model as one with superior performance in modeling the outcome variables of COVID-19 infections in counties within the United States, outperforming KNN and Decision Trees by having better accuracy and Kappa values.



3. Evaluation

The usefulness of the Random Forest model for stakeholders, especially health departments and policymakers, is evident as it has proven to yield a high degree of Kappa value regarding the prediction of COVID-19 outcomes. The model allows the determination of counties at risk of serious outbreaks and aids in intervention by analyzing major variables that include death percentages, infection rates, and population health factors.

Usefulness to Stakeholders:

These would include government health departments, local health authorities, and policymakers who can then use these predictions to undertake necessary and timely actions. Specifically, the model could help identify areas where the virus is likely to spread rapidly or where healthcare resources are insufficient. In turn, these insights can inform the following decisions:

Resource Allocation: Predict where additional healthcare resources- such as hospital beds, ventilators, and medical supplies-should be sent.

Targeted Lockdowns and Restrictions: Based on predicted infection rates and current

health care capability, recommend possible counties that may need stronger containment.

Vaccination Distribution: Prioritize counties with higher predicted risks for vaccine rollouts.

4. Deployment-

The Random Forest model would be deployed as a decision-support tool for public health authorities. It would allow these stakeholders to predict the trajectory of COVID-19 outbreaks in real time, making it possible to allocate resources more efficiently and implement targeted interventions.

Practical Use:

The model will be integrated into a real-time dashboard showing the predicted COVID-19 trends for each county. This can be integrated with live data feeds from health departments, census data, and other public health sources to give current predictions.

Actionable insights of the model will include:

Risk Zones: Identifying counties with the highest likelihood of seeing a spike in cases and deaths.

Healthcare Readiness: Predicting which regions need additional healthcare support, such as ICU beds, medical staff, and supplies.

Lockdown/Intervention Predictions: Advising on whether counties need stricter movement restrictions or social distancing measures.

Actions Based on Model's Output:

Prioritize Testing and Vaccination: Direct resources to high-risk areas based on model predictions.

Policy Adjustments: Inform government decisions about lockdowns, travel restrictions, and other measures to control the spread of viruses.

Ongoing Monitoring and Adjustment: Regularly monitor the model performance and adjust interventions based on updated predictions.

Model Updates: The model needs to be updated on a routine basis, if possible, on a weekly or bi-weekly basis, as the conditions change with the new data. Some key factors driving the need for updates are:

New COVID-19 Variants: The model should include data on new variants and their possible consequences on infection rates and severity as the virus evolves.

Shifts in Public Health Policy: Changes in vaccination strategies, lockdown measures, and other government actions should be factored into the model's predictions. The model should also consider changes in the population, such as population growth or migration patterns. Continuous Learning:

One possible improvement that could be made to the model is the addition of online learning, whereby it can learn and update its predictions with incoming data. This will make the model more adaptive and ensure it stays relevant with emerging challenges.

5.

Appendix

Resources

Project starter code

https://mhahsler.github.io/Introduction_to_Data_Mining_R_Examples/book/classification-alternative-techniques.html#model-comparison-1

6. Exceptional Work -

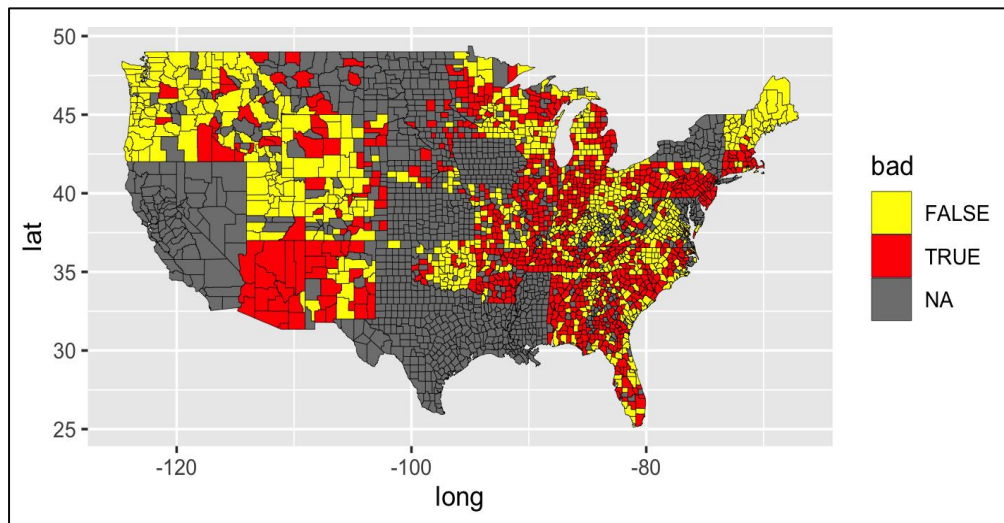
6.1 Classification Using Naive Bayes Models

The Naive Bayes (NBfit) classification model uses a probabilistic technique to categorize U.S. counties as "Endanger" (red) or "Not Endanger" (yellow) based on COVID-19 death percentages that exceed the threshold of 0.11%. The ground truth map shows how the real county classification process was done: safer counties are shown by yellow, whereas high-risk counties with a fatality rate above 0.11% are represented by red. On the other hand, the model's predictions are displayed in the NBfit predictions map, which replicates the classification according to the same condition. A visual comparison of these maps reveals both areas of misconceptions and significant alignment zones.

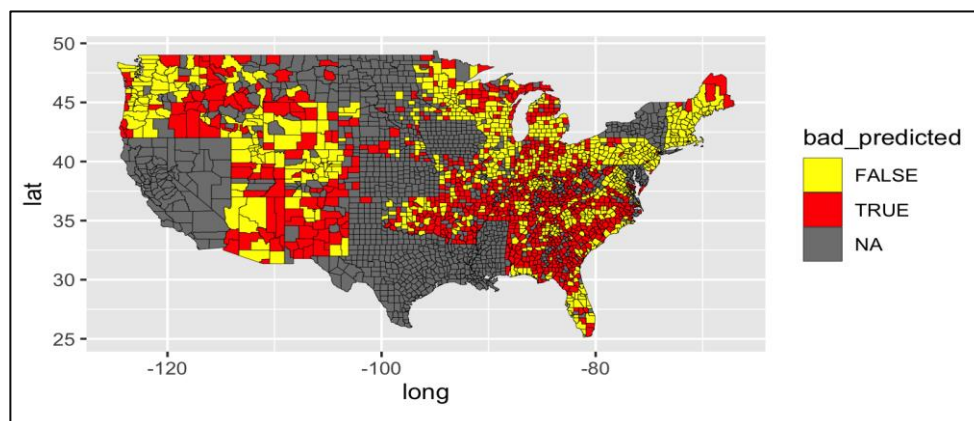
Visual analysis of the maps reveals a high degree of agreement between the ground reality and the model's predictions, especially in areas with large fatality rate clusters. The model's effectiveness in identifying these trends was confirmed by the fact that dangerous counties were mostly found in regions that had previously seen higher COVID-19 mortality. Disparities do exist, though, especially when it comes to overestimation of risk (false

positives) and underestimation (false negatives). These mistakes highlight how crucial it is to take into account county-level particulars, like healthcare capacity, socioeconomic vulnerabilities, and movement patterns, in order to improve future forecasts.

Ground Truth

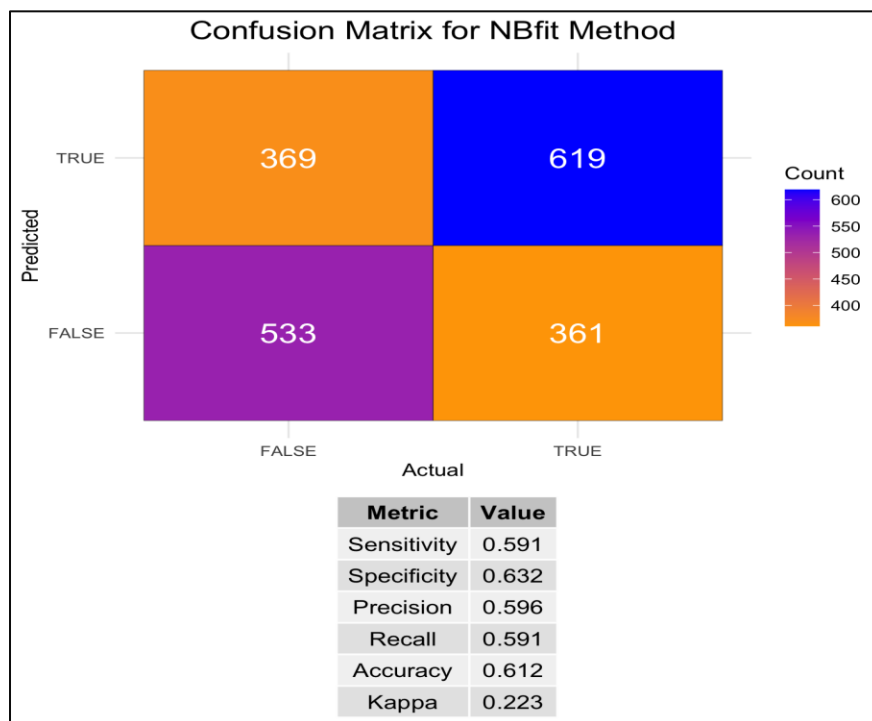


Predictions



The performance of the NBfit model was done with the confusion matrix, which showed an

accuracy of 61.2%, meaning that about 61% of counties were correctly classified. The sensitivity, measuring the model's ability to identify high-risk counties correctly, was 59.1%, while the specificity, representing its capacity to identify safe counties correctly, was slightly higher at 63.2%. This means that the model performs marginally better in classifying safer counties. The precision for high-risk classifications is 59.6%, implying that about 60% of the counties classified as being at high risk were correct. Similarly, the Kappa statistic with a value of 0.223 showed moderate agreement between the predictions against the ground truth and was indicative of further improvements needed to ensure better capturing of county characteristics.



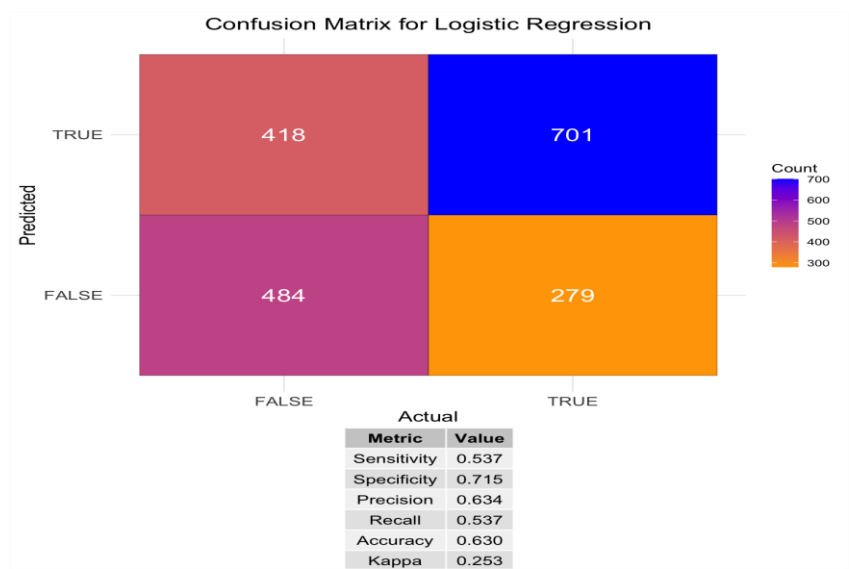
6.1 Classification Using Logistic Regression model

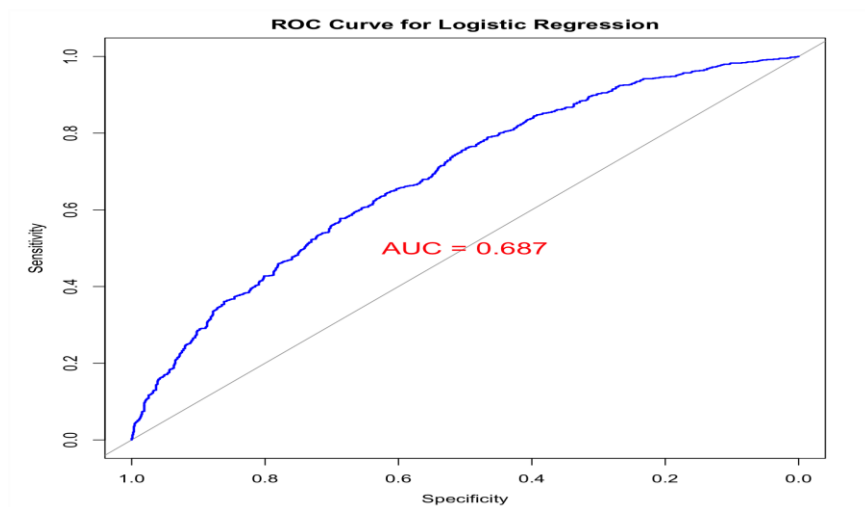
Overview

Logistic Regression is a commonly used statistical classification strategy for binary outcomes. Based on socioeconomic, demographic, and mobility-related statistics, it was used in this effort to categorize U.S. counties into "high risk" or "low risk" groups. The goal is to use these findings to identify counties that would need early interventions and improved public health infrastructure in order to prepare for future pandemics. Because of its simplicity and interpretability, logistic regression was chosen as a potent technique for

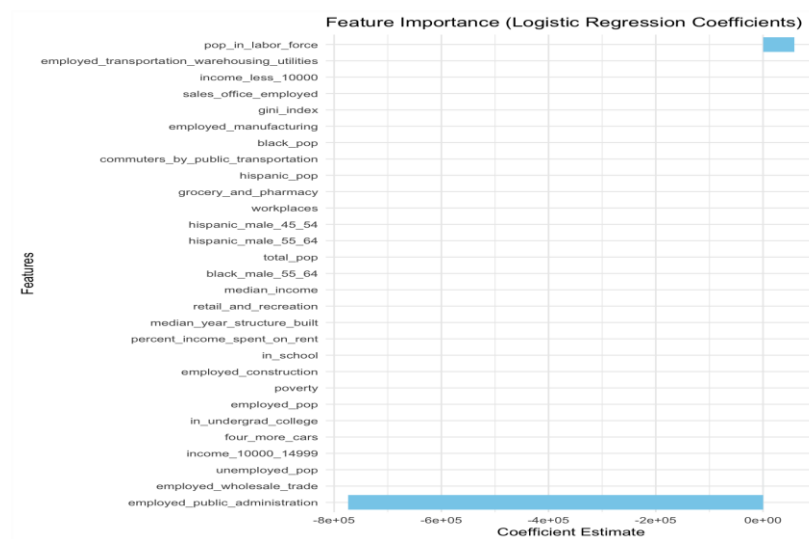
extracting useful information from complicated datasets. In addition to the projected classifications, the model's outputs contain the probability of being categorized as "high risk," which were visually represented to help identify regional trends.

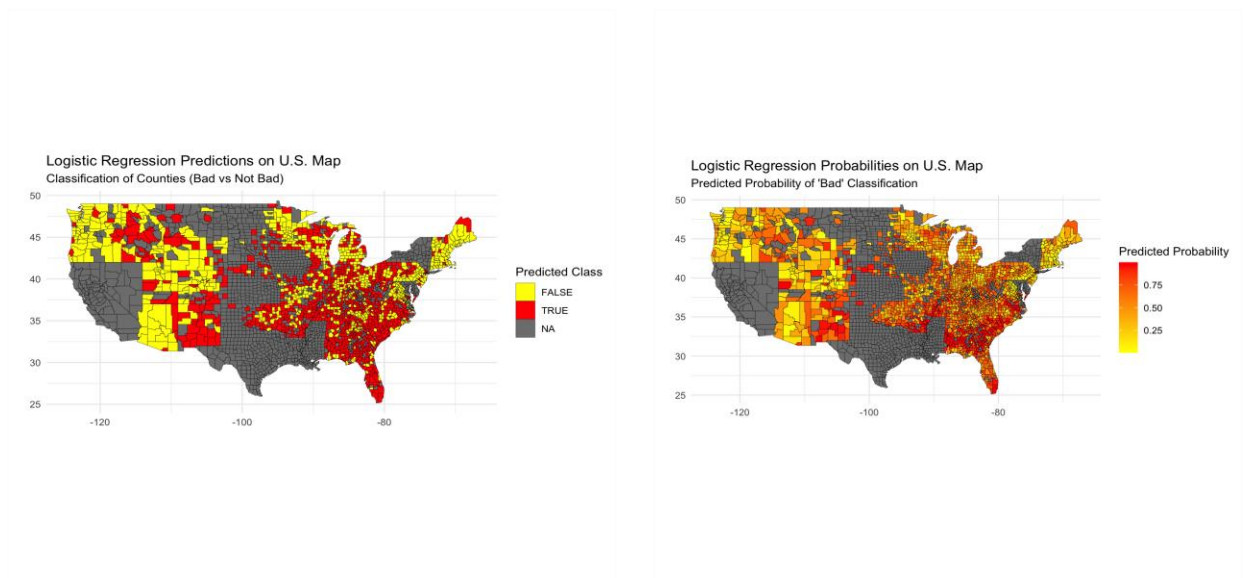
A thorough analysis of the Logistic Regression model's classification performance may be found in the confusion matrix. Counts for false positives (484), false negatives (279), true positives (701), and true negatives (418) are shown in the matrix. Understanding how successfully the model separates counties categorized as "high-risk" (TRUE) and "low risk" (FALSE) depends on this image. The comparatively high specificity (71.5%) demonstrates the model's strength in accurately identifying low-risk counties, even though the total accuracy of 63% suggests space for improvement. This is essential to the project's objective since it guarantees that funds are not misdirected to counties that don't require urgent assistance. Additionally encouraging is the precision score of 63.4%, which shows that the model is accurate in predicting a high-risk county in a significant amount of the time





The trade-off between specificity and sensitivity (recall) at different categorization thresholds is displayed by the ROC curve. The Logistic Regression model's fair ability to differentiate between high-risk and low-risk counties is indicated by its area under the curve (AUC) of 0.687. Given that the AUC offers a comprehensive assessment of the model's performance, this is especially pertinent to the project's objective of getting ready for potential pandemics. The model's ability to efficiently balance false positives and false negatives is indicated by the smooth curve, which is crucial for identifying counties that might require early interventions without overburdening resources. The model's performance might be greatly enhanced with possible feature engineering and hyperparameter adjustment, and the AUC number provides a strong basis for more improvement.





Understanding which factors have the greatest influence on the model's predictions is made possible by the feature importance plot that is produced from the Logistic Regression coefficients. Employed_public_administration, the most notable attribute, has a dominant coefficient, suggesting a substantial correlation with pandemic outcomes. This supports the theory that during a pandemic, public administration and associated infrastructure are essential for risk management and mitigation. Other characteristics, such as the number of people in the labor force and income levels, also offer insightful information, emphasizing employment and socioeconomic factors as important determinants. In addition to helping interpret the model, this visualization directs policymakers to concentrate on enhancing employment and public health infrastructure in order to reduce risks in the future.

A geospatial summary of the model's classifications is given by the U.S. map of projected classes, which displays counties categorized as low-risk (yellow) or high-risk (red). The initiative will benefit greatly from this visualization since it immediately aids in decision-making by pointing out regions that might need specific actions, such as mask laws or company closures. The model's predictions are further validated by the spatial distribution of forecasts, which shows patterns consistent with established demographic and socioeconomic trends. For public health professionals and legislators to efficiently prioritize resources, this map is a crucial tool. An additional level of analysis is provided by the U.S. map of anticipated probabilities, which shows the probability that each county will

be categorized as high-risk. Instead of using a one-size-fits-all strategy, the gradient from yellow (low probability) to red (high probability) enables a nuanced assessment of risk levels and customized responses. For instance, counties with high likelihood can be given priority for quick action, while those with intermediate probabilities might be watched more intently. This map highlights the value of the Logistic Regression model in planning and pandemic preparedness by showcasing its ability to produce meaningful information.

6.2. Evaluation:

Naive Bayes (NBfit) and Logistic Regression, the categorization models created for this research, give stakeholders vital tools to lessen the effects of possible future pandemics. Based on socioeconomic, demographic, and mobility-related factors, the models divide U.S. counties into high-risk and low-risk zones using data from Projects 1 and 2. These include indicators like income levels, public administration employment, workplace activity and other mobility trends, and the percentages of Black and Hispanic people. Particularly in areas with significant vulnerabilities, these models provide a basis for developing focused responses, such as allocating healthcare resources, implementing mask requirements, and starting temporary closures.

Using COVID-19 death rates over a threshold of 0.11%, the Naive Bayes model classified counties in several states, including California (West), Mississippi (South), and New York (Northeast), as "Endanger" (high-risk) or "Not Endanger" (low-risk). When the model's projections and ground truth are closely compared, areas with notable COVID-19 fatality clusters, such as the Midwest (South Dakota, Iowa) and the Deep South (Mississippi, Alabama), show alignment. Although certain false positives (such as areas in the Great Plains) and false negatives (such as portions of California's Central Valley) were noted, the model's probabilistic foundation enabled it to successfully identify these trends. These discrepancies imply that for future apps to be more accurate, geographical aspects like social distancing compliance and healthcare accessibility must be taken into account.

The confusion matrix shows 61.2% accuracy, 59.1% sensitivity, and 63.2% specificity. The model did slightly better at identifying low-risk counties, like those in the Mountain West region (e.g., Utah and Nevada), according to the slightly greater specificity. The sensitivity

score, however, indicates that there is potential for improvement in detecting high-risk areas, particularly in states with significant population densities like Texas and California. Regions like Mississippi and Louisiana, where socioeconomic vulnerabilities and healthcare inequities cross, are excellent candidates for early interventions during future outbreaks, and the model's performance is especially helpful in identifying these locations.

The Logistic Regression model focused on counties in Texas, New York, Maryland, and Kansas, providing insights into risk levels in a variety of geographic and socioeconomic circumstances. High-risk classifications (TRUE), for example, were concentrated in metropolitan regions such as New York City, where population density and public health infrastructure are important factors in pandemic response. The model's precision (63.4%) and specificity (71.5%) show that it is dependable in accurately recognizing low-risk counties (like Kansas' rural areas) while retaining a high degree of accuracy in high-risk classifications. A more sophisticated understanding of how socioeconomic and mobility characteristics contribute to pandemic risk was made possible by the inclusion of variables like retail mobility and employment in public administration. Regional patterns, such the concentration of high-risk counties in the Northeast (like New York and New Jersey) and the South (like Mississippi and Alabama), are highlighted in the maps produced from Logistic Regression predictions and probabilities. The Rio Grande Valley in Texas and the Central Valley in California, for instance, have intermediate probabilities, indicating the necessity of careful observation and gradual resource distribution. Furthermore, the feature importance analysis emphasized the importance of variables connected to employment, highlighting the necessity of labor dynamics in forecasting the impact of a pandemic. This data can help stakeholders deprioritize low-risk rural counties in the Midwest and concentrate on vulnerable areas, including California's metropolitan centers.

Regional and Variable-Specific Insights

Evaluations of the models provide important information about vulnerabilities that are specific to different regions and variables. For instance:

Western States (California, Nevada): Due to socioeconomic disparities and mobility factors, high probability was noted in metropolitan counties such as Los Angeles and Fresno.

Southern States (Mississippi, Alabama): Racial inequities in pandemic preparedness must be addressed because high-risk areas were concentrated in low-income areas with sizable Black populations.

Northeastern States: Employment in public administration and urban density were found to be important determinants, emphasizing the necessity of strong public health systems in these areas (New York, New Jersey).

Midwestern States (Kansas, Iowa): The models' capacity to differentiate between high- and low-risk areas was validated by the accurate identification of low-risk counties with steady socioeconomic indicators.

Governments can create specialized interventions by combining this region-specific information, such as strengthening healthcare facilities in the Deep South or enforcing more stringent movement restrictions in urbanized regions during epidemics. These assessments confirm the models' usefulness in anticipating pandemics in the future and guaranteeing fair resource distribution among various geographical areas.

6.3. Deployment

The use of the Naive Bayes and Logistic Regression models would be critical to improve pandemic preparedness by allowing for data-driven decision-making. Based on socioeconomic, demographic, and mobility criteria, these models would divide U.S. counties into high-risk and low-risk groups, giving policymakers useful information. Their forecasts may direct focused responses, such as giving high-risk counties—like urban areas in California and New York—priority when allocating healthcare resources, launching immunization campaigns, and improving infrastructure. Customized preventative actions that are proportionate to their needs would be beneficial for counties with medium to low-risk classifications, such as some rural areas in Kansas or Iowa. This strategy minimizes the spread of upcoming pandemics by ensuring that scarce resources are used effectively. In practical terms, federal and state authorities would use these models as part of a centralized pandemic response system. During epidemics, they would enable prompt action by powering live dashboards that offer real-time updates on county risk levels. This information could be used by policymakers to enact prompt policies like mask requirements,

mobility limitations, and focused public health initiatives. Furthermore, the probabilistic results of the models would enable stakeholders to assess the level of risk, supporting more nuanced decision-making as opposed to a one-size-fits-all strategy. For example, counties with excessive danger probability would be immediately intervened in, whereas counties with relatively high-risk probabilities may be continuously monitored.

The models would be updated regularly to preserve their relevance and accuracy. Data inputs including mobility trends and healthcare capacity measurements would be updated every week during active pandemic phases, and complete model retraining would take place every month. To guarantee preparedness for upcoming crises, updates would take place every three months during non-pandemic times. By addressing changing variables, this ongoing improvement would guarantee that the models continue to be reliable and flexible. These models would greatly improve public health authorities' capacity to anticipate, prepare for, and effectively respond to future pandemics by adjusting their outputs in real-time to reflect changing demands.